THE RESOLUTION PROPERTY DESCENDS ALONG SURJECTIVE FINITE LOCALLY FREE MORPHISMS

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Allowed standard prerequisites (editorial).

- A finite locally free morphism is affine and its direct image functor on quasi-coherent modules is exact, by the relative affine vanishing lemma. invoked at proof lines 9529-9535. Provides the exact surjection after applying f_*.
- The projection formula f_* f^* G = G tensor f_* O_X for the affine morphism. invoked at proof lines 9536-9541. Transfers the surjection from X to Y.
- For a ring map A -> B finite locally free and a finite locally free B-module N, the A-module N is finite locally free, via the algebra lemmas on finite projective modules, finite presentation under extension, and composition of flat maps. invoked at proof footnote lines 9547-9562. Justifies that f_* E is finite locally free over Y, so the dual surjection exhibits G as a quotient of a finite locally free module.

Source reference. Original lemma 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cited at proof lines 9534-9535 for the exactness of f_* on quasi-coherent modules; the pushforward of the surjection is inside the counted span.

Source reference. Original lemma 00NX: sources/stacks-chow/tags.txt, source lines 875–875. Cited at proof lines 9549-9562 twice inside the footnote descent of finite local freeness; the footnote is inside the counted span.

Source reference. Original lemma 0564: sources/stacks-chow/tags.txt, source lines 6697–6697. Cited at proof footnote lines 9556-9557 of the descent chain; the footnote is inside the counted span.

Source reference. Original lemma 00HC: sources/stacks-chow/tags.txt, source lines 638–638. Cited at proof footnote line 9558 of the descent chain; the footnote is inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a surjective finite locally free morphism of schemes. If X has the resolution property, so does Y.*

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Proof. The condition means that f is affine and that $f_*\mathcal{O}_X$ is a finite locally free $\mathcal{O}_Y$-module of positive rank. Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_Y$-module of finite type. By assumption there exists a surjection $\mathcal{E} \to f^*\mathcal{G}$ for some finite locally free $\mathcal{O}_X$-module $\mathcal{E}$. Since f_* is exact on quasi-coherent modules (Cohomology of Schemes, Lemma 01XC) we get a surjection

$$f_*\mathcal{E} \longrightarrow f_*f^*\mathcal{G} = \mathcal{G} \otimes_{\mathcal{O}_Y} f_*\mathcal{O}_X$$

Taking duals we get a surjection

$$f_*\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{O}_X, \mathcal{O}_Y) \longrightarrow \mathcal{G}$$

Since $f_*\mathcal{E}$ is finite locally free[1], we conclude. □

[1]Namely, if $A \to B$ is a finite locally free ring map and N is a finite locally free B-module, then N is a finite locally free A-module. To see this, first note that N finite locally free over B implies N is flat and finitely presented as a B-module, see Algebra, Lemma 00NX. Then N is an A-module of finite presentation by Algebra, Lemma 0564 and a flat A-module by Algebra, Lemma 00HC. Then conclude by using Algebra, Lemma 00NX over A.